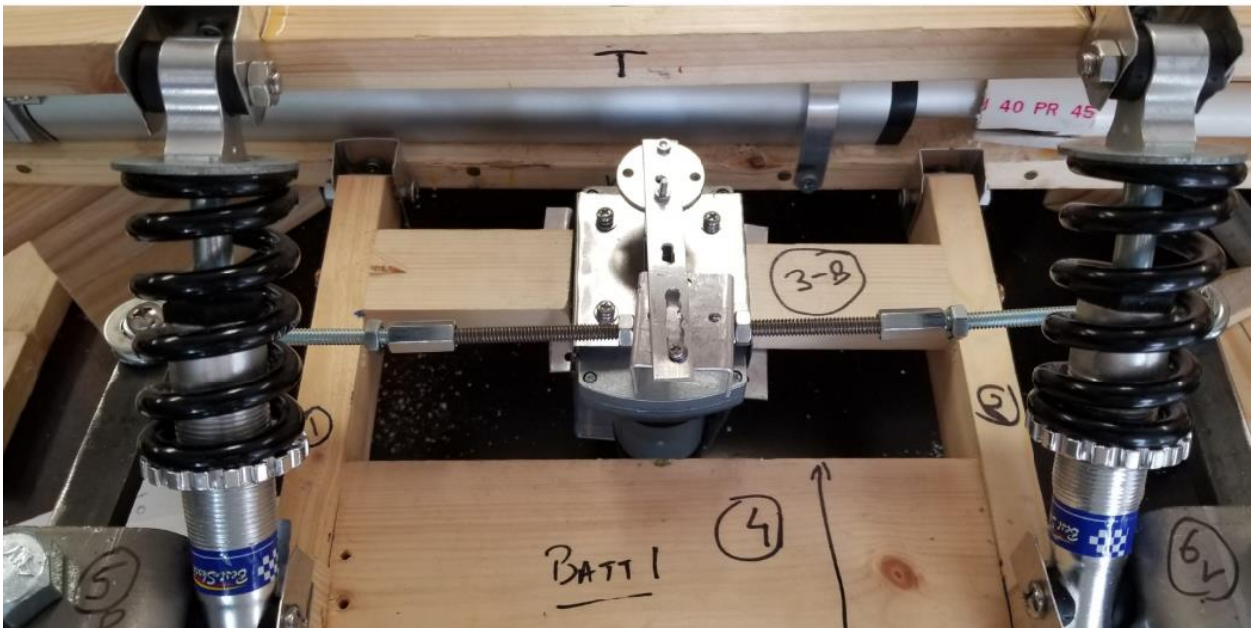
1. **What is the exact current requirement for the Servo? The suggested part in document has a rating of 160A. Is this inline with the requirements?**

   I think 160 A is too high. Please see the attached spec sheet 34108 MDData – it states 500mA – No Load. I have this motor on my bench if you want I can check the current requirements using a multimeter. The purpose of this servo is to control the rack and pinion function of the cart. Here is a picture of motor installed on a clay model. I think we should keep the output voltage to 12 volts for this motor.



2. **What is the purpose of Linear Actuator / Solenoid? The controlling is not a problem. I was just curious about the purpose.**

   See attached video. The Linear Actuator is to open and close ramps where wheelchairs can be loaded and unloaded from.

   The solenoid will be used to open doors that are controlled by remote – rather than programming universal door opener in the system – We think it will be easier to allow the user to install the door remote in a box on the cart and the solenoid will serve as a clicker (same as finger) to open and close handicap assisted doors by tapping on the remote button.

3. **What should be the rating of the battery? I get that we need a 12V battery to power up the Board. Have you done any current calculations yet to determine what capacity is required to meet the 2hrs functional requirement? Or should I carry out the assumptions of current?**
   We were planning on using something simple like a ML18-12 is a 12V 18AH sealed lead acid (SLA) rechargeable maintenance free battery

Link: https://www.lowes.com/pd/Mighty-Max-Battery-Mighty-Max-SLA-Rechargeable-Sealed-Lead-Acid-12180-Backup-Power-Batteries/1001107752?cm_mmc=shp-_-c-_-prd-_-hdw-_-google-_-pla-_-214-_-sosbatteries-_-1001107752-_-0&placeholder=null&gclid=Cj0KCQjwgJv4BRCrARIsAB17JI6C5psmTFVU2GCo7TEx8Zdu2iSDTlXxvTKAcj3yABoI5_ldZgWnBJwaAhz5EALw_wcB&gclsrc=aw.ds

If you think this is not sufficient, we can switch to LiPoly batteries. If you think that the battery will not be able to supply 2 hours of on time and 1-hour continuous run please suggest what should we use?

4. **What should be the resolution / characteristic of the camera? I just had a good idea for interfacing the camera which I'd like to research upon after I know the requirements.**

For resolution we were thinking WXGA (1280x800 pixels, 16:10 aspect ratio) but if there is an alternative which is higher resolution, we are fine with that, anything lower like a vga may not work. The remotes layout should be like Spot robots remote: here is an image and I think WXGA or higher may satisfy the need.

The layout of the buttons will be as in the drafting requirements document but this is benchmark that we want to strive for.



